Edge-graceful labellings of odd-order trees with few vertices of degree two

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Abstract

A tree T of odd order n is edge-graceful if its edges can be labelled bijectively by the nonzero residues modulo n so that the vertex sums are all the residues modulo n. Lee conjectured in 1989 that every tree of odd order is edge-graceful. The case of trees with at most one vertex of degree two follows from the zero-sum partition theorem of Kaplan, Lev and Roditty, and we recently settled the case of at most two such vertices. We prove that there are absolute constants C, A and n_0 such that every tree of odd order $n \geq n_0$ with D vertices of degree two is edge-graceful whenever $n > C(D+1)(\log n)^A$; in particular, for every fixed k, every large enough tree of odd order with at most k vertices of degree two is edge-graceful. The proof combines a finite table of local labelling identities, found by an exact constraint search and re-verified by an independent checker, with a bottom-up scheduling argument and with the zero-sum matching theorem of Müyesser and Pokrovskiy. Under the abelian improvement of that theorem which its authors record in their concluding Section 7, the threshold becomes linear, $n > C(D+1)$. The constants are not effective. We also reduce the linear threshold to an elementary statement about permutations of an interval, which we prove in several cases and verify exactly up to order 30, and which would make the constant explicit.

1 Introduction

Let T be a tree with n vertices, n odd. An *edge-graceful* labelling of T is a bijection $\lambda\colon E(T) \to \mathbb{Z}_n \setminus \{0\}$ such that the induced vertex sums $s(v) = \sum_{e \ni v} \lambda(e)$, taken modulo n, form a bijection $V(T) \to \mathbb{Z}_n$. Lo [5] introduced the notion and Lee [4] conjectured that every tree of odd order is edge-graceful. The parity condition is necessary, because the sum of all edge labels is $\binom{n}{2}$, which is $\equiv 0$ modulo n only for odd n. The conjecture is surveyed in Gallian's dynamic survey [2]. The strongest general result we are aware of is due to Kaplan, Lev and Roditty [3], who proved that $\mathbb{Z}_n \setminus \{0\}$, n odd, admits a zero-sum partition into parts of any prescribed sizes at least two, and deduced that every odd-order tree with at most one vertex of degree two is edge-graceful. Their argument identifies the obstruction exactly, since a vertex of degree two would require a zero-sum block of size one, which is impossible. In [6] we extended their theorem to trees with at most two vertices of degree two, by combining zero-sum partitions with perfect and hooked Langford sequences and a case analysis verified computationally for all odd $n \leq 5001$. The argument there is specific to the two degree-two vertices and does not extend to three.

In this paper we show that the number of degree-two vertices, rather than their placement, is what limits the zero-sum approach, provided that it is small compared with the order of the tree.

Theorem 1.1. *There are absolute constants C, A and n_0 such that every tree T of odd order $n \geq n_0$ with D vertices of degree two is edge-graceful whenever $n > C(D+1)(\log n)^A$. In particular, for every $k \geq 0$ there is an n_k such that every tree of odd order $n \geq n_k$ with at most k vertices of degree two is edge-graceful.*

The cases $k = 0$ and $k \leq 2$ of the second statement are the theorems of [3] and [6], respectively, with $n_k = 1$ there. The constants come from an asymptotic matching theorem and are not effective, and the theorem says nothing about trees in which the degree-two vertices have positive density, e.g. caterpillars with many subdivided edges, the regime of the full conjecture.

The only external input is a zero-sum matching theorem in $\mathbb{Z}_n$ that tolerates a set of deleted residues, and it is the size of that set, not the matching itself, that determines the threshold. The printed statements of Müyesser and Pokrovskiy [7] tolerate $n/(\log n)^A$ deletions and give Theorem 1.1. However, their Section 7, under *Bounds in the main theorem*, records that for abelian groups the logarithms may be removed, since the only source of a logarithm in their proof is a bound on the length of a commutator product and that product is empty in an abelian group; the authors state this improvement and use it themselves, although it is not a printed theorem. Under that reading our argument gives a linear threshold.

Theorem 1.2. *Assume the abelian improvement recorded in [7, Section 7,* Bounds in the main theorem*], namely that their zero-sum matching theorems hold in $\mathbb{Z}_n$ with a linear number of deleted residues. Then there are absolute constants C and n_0 such that every tree T of odd order $n \geq n_0$ with D vertices of degree two is edge-graceful whenever $n > C(D+1)$. In particular, for every $k \geq 0$, every odd-order tree with at most k vertices of degree two and more than $C(k+1)$ vertices is edge-graceful, since $D \leq k$.*

The qualitative statement, that a bounded number of degree-two vertices is no obstruction once the tree is large, therefore does not depend on that remark at all; the remark is what turns the polylogarithmic threshold into a linear one. In Section 7 we reduce the linear threshold, with an explicit constant and without any use of [7], to a conjecture on permutations of an interval that we prove in several cases and verify exactly up to order 30.

Our approach separates the problem into a local part and a global part. Locally, each maximal path of degree-two vertices, together with the branch vertex that owns it, is labelled by an explicit family of integer forms whose vertex sums permute its own labels (we call this the $P = O$ property). Such families are found by our exact search over all output permutations and constraint programming, and every family used in the proof is re-verified by a program that shares no code with the search. Globally, the families are assembled bottom-up so that the multiset of special labels is closed under negation up to at most one zero-sum triple and has size and magnitude linear in D. The remaining labels are then partitioned into zero-sum blocks of the prescribed sizes by the random Hall–Paige theorem of Müyesser and Pokrovskiy [7]. Therefore the local part is a finite computation (12,742 certified families), whereas the global part is a scheduling lemma, whose regression is an executable scheduler that we tested on all trees with at most nine vertices and on thousands of random trees.

The paper is organised as follows. Section 2 reformulates edge-gracefulness as a permutation identity and fixes the vocabulary. Section 3 decomposes the tree and proves the counting lemmas. Section 4 describes the local families and the finite certificate table. Section 5 proves the sequential realization lemma. Section 6 applies the dense completion and proves Theorems 1.1 and 1.2. Section 7 records the elementary route to the linear threshold. Section 8 describes the computations and the certificate format.

2 Set-up

Throughout, T is a tree of odd order n rooted at a vertex ρ of degree at least three (if T is a path the explicit labelling $1, 2, \ldots, n-1$ along the path is edge-graceful, so this case is excluded). For $v \neq \rho$ let e_v denote the parent edge of v and put $\lambda(e_\rho) = 0$. A vertex is *stationary* if the labels of its child edges sum to zero, so that $s(v) = \lambda(e_v)$. Let W be the set of non-stationary vertices together with ρ and let $S = \{\lambda(e_w) : w \in W \setminus \{\rho\}\}$, the *special support.*

Lemma 2.1. *λ is edge-graceful if and only if the multiset $\{s(w) : w \in W\}$ equals $\{0\} \cup S$.*

Proof. For a stationary vertex $v \neq \rho$ the sum $s(v)$ equals the label of e_v, so the stationary vertices contribute exactly the labels of their parent edges. The remaining labels are $\{0\} \cup S$ and they must be produced by W. □

Every vertex of degree two is in W, because a single nonzero child label cannot sum to zero. Therefore our construction chooses W of size $O(D)$, arranges the identity of Lemma 2.1 cell by cell, and leaves every other internal vertex a zero-sum block of size zero or at least two. We work with integer labels of magnitude $O(D)$ and reduce modulo n at the end.

Core and arms. Suppressing the degree-two vertices of T gives the core Q, a tree whose vertices have degree one or at least three in T, and a core edge carries a length ℓ, the number of suppressed vertices. We process Q bottom-up. A subdivided child edge of a core vertex v is an *arm* of v when its lower endpoint is not an owner (defined below), and the *head chain* of the lower endpoint otherwise. A direct child edge whose lower endpoint is not an owner is a *port.*

Cells and $P = O$. A *cell* at a core vertex v consists of the labels of v's parent edge (or of its head chain), of its residual arms and of some recruited ports, and possibly of the edge to one or two active children. Its *physical multiset* P is the multiset of these labels. Its *output multiset* O consists of the sum at v, the adjacent sums along each arm and chain, the terminal label of each arm, and the label of each port. A cell is *self-permuting*, written $P = O$, when the two multisets coincide, and for a root cell the constant 0 is adjoined to P. Substituting a $P = O$ cell for a stationary child edge preserves the identity of Lemma 2.1 because the shared edge is counted once on either side.

3 Decomposition and counting

3.1 Owner-free neutralization

Lemma 3.1 (neutral units)**.** *Two arms whose lengths have the same parity and whose residues modulo six are not $\{1, 3\}$, and the four-arm multisets with residues $(1, 1, 1, 3)$ and $(1, 3, 3, 3)$, admit labellings whose adjacent sums and terminals permute their own labels, whose first labels sum to zero, and whose support is closed under negation, and each has two free integer parameters. Two arms of lengths one and three admit no such labelling.*

The thirteen families for the residue representatives are found by our constraint search of Section 4 (a root row with pinned zero output) and re-verified by the independent checker. They extend to all lengths in the residue classes by the six-edge insertion, which lengthens an arm by six while adding three inverse pairs, and the impossibility for lengths one and three is an exact enumeration. We call these labelled arm sets *neutral units.* (Pairs whose residues are $\{1, 3\}$ but whose lengths are not, e.g. $(1, 9)$ or $(3, 7)$, also admit such labellings by the same search. We do not use them, since a pair of this type is carried instead in the residual, which has families of its own.)

Corollary 3.2 (six residual types)**.** *The arm multiset of every core vertex can be partitioned into neutral units and a* residual *of one of six types, namely empty, one odd arm, one even arm, a pair with residues $\{1,3\}$, one odd and one even arm, or such a pair together with one even arm.*

Proof. Pair the even arms arbitrarily, leaving at most one even arm. Among the odd arms, pair every arm of residue five with any other odd arm, then pair arms of residue one with each other and arms of residue three with each other. What remains is at most one arm of residue one and at most one of residue three. Thus the odd part of the residual is empty, a single odd arm, or a pair with residues $\{1,3\}$, and the even part is empty or a single arm. □

3.2 Owners and contexts

Definition 3.3. A nonroot core vertex is an *owner* if its residual is nonempty, or if exactly one active child remains after the cancellations described below and no port is available to absorb it. A vertex with empty residual and no remaining active child is stationary, and its parent edge, if subdivided, is an arm of its parent.

At a vertex with $k \geq 2$ active children (children which are owners) the heads are free parameters of the children's cells, so they are prescribed to sum to zero (a mirror pair for $k = 2$, a free zero-sum triple for $k = 3$, pairs and at most one triple otherwise), and none remains. A single remaining active child with head x is absorbed by a stationary vertex with q ports by a helper port labelled $-x$ ($q = 1$ or $q \geq 3$) or by two ports labelled a and $-x - a$ ($q = 2$). Two exceptional child types cannot be absorbed in this way, because every $P = O$ family of the child contains the antipode of its own head: the residue-one letter and the $(1; \text{one port})$ bottom, and such children are joined into their parent's cell.

The *context* of an owner is the tuple (h, R, p, c) where h is the length of its head chain reduced modulo six, R its residual arm multiset with lengths reduced modulo six, $p \in \{0, 1, 2, 3\}$ the number of recruited ports (never leaving exactly one unrecruited), and c the child type, one of *none*, *free*, the two joined types, or the two-input type used when two active heads are kept. The root has context (root, R, p, c).

Reduction modulo six is justified by the following identity, which lengthens one arm of a cell by six and changes nothing else.

Lemma 3.4 (six-edge insertion)**.** *Let an arm of a cell carry the labels $a_0, \ldots, a_\ell$, a_0 at the owner, so that it contributes the outputs $a_0 + a_1, \ldots, a_{\ell-1} + a_\ell$ and a_ℓ. Write $c = a_0$ and $d = a_1$ and insert the six labels*

$$\tfrac{-2c-d}{3}, \quad \tfrac{c-d}{3}, \quad \tfrac{c+2d}{3}, \quad \tfrac{2c+d}{3}, \quad \tfrac{-c+d}{3}, \quad \tfrac{-c-2d}{3}$$

between a_0 and a_1. They form three pairs of opposite forms, and the seven outputs they create are exactly those six labels together with the output $c + d$ that the arm had before. For a port, an arm of length zero whose only label is c and whose output is c itself, the same holds with the six labels

$$\tfrac{c}{3}, \quad \tfrac{-2c}{3}, \quad \tfrac{4c}{3}, \quad \tfrac{-c}{3}, \quad \tfrac{2c}{3}, \quad \tfrac{-4c}{3}$$

and with c in place of $c + d$.

Proof. For the first list the outputs are, in order, $c + \frac{-2c-d}{3} = \frac{c-d}{3}$, $\frac{-2c-d}{3} + \frac{c-d}{3} = \frac{-c-2d}{3}$, $\frac{c-d}{3} + \frac{c+2d}{3} = \frac{2c+d}{3}$, $\frac{c+2d}{3} + \frac{2c+d}{3} = c + d$, $\frac{2c+d}{3} + \frac{-c+d}{3} = \frac{c+2d}{3}$, $\frac{-c+d}{3} + \frac{-c-2d}{3} = \frac{-2c-d}{3}$ and $\frac{-c-2d}{3} + d = \frac{-c+d}{3}$, which is the asserted multiset, and the three pairs are the first with the fourth, the second with the fifth and the third with the sixth. The port list is checked the same way. □

Thus a family for a context is also a family for the context with any one arm six longer, and in particular a port may be replaced by an arm of length six. The new labels are forms in the two labels the insertion sits between, so the identity is one statement about all cells at once. The insertion may make two labels of the cell equal for special values of c and d, which the search rules out by requiring the label forms to stay pairwise distinct. A twelve-label endpoint preserving insertion does the same for head chains.

A second identity removes a much larger part of the search. Most of the contexts of the list differ from a narrower one only by a set of arms and ports that can be made to pay for itself, and such a set may be adjoined to any family at all.

Definition 3.5. A *block* is a multiset of arm lengths $\ell_1, \ldots, \ell_k$ together with a number $m \geq 0$ of ports, and it carries $N = m + \sum_i (\ell_i + 1)$ labels. The block is *neutral* if there are integers $c_1, \ldots, c_N$ such that, when its labels are taken to be $c_1 t, \ldots, c_N t$ in a single parameter t,

(N1) the labels it contributes to the owner, namely its m port labels and the first label of each of its arms, sum to zero;

(N2) the multiset of its N labels equals the multiset of the N outputs at the vertices it introduces, that is the m port leaves and, for an arm $a_0, \ldots, a_\ell$, the outputs $a_0 + a_1, \ldots, a_{\ell-1} + a_\ell$ and a_ℓ;

(N3) the multiset $\{c_1, \ldots, c_N\}$ is closed under negation; and

(N4) the c_i are nonzero and pairwise distinct.

It is a *token block* if it satisfies (N1), (N2) and (N4) and, instead of (N3), three of its labels sum to zero and span a plane while the remaining $N - 3$ are closed under negation. A token block therefore needs at least two parameters, and N is odd for it and even for a neutral block.

Which of the two a given set of arms and ports can be is decided by parity alone. Two arms of lengths a and b carry $a + b + 2$ labels, so a same-parity pair can only be a neutral block and a pair of opposite parity can only be a token block, which is the arithmetic behind the neutral units of Section 3: an $\{o, e\}$ residual is exactly the residual that forces a token.

Lemma 3.6 (neutral adjunction)**.** *Let B be a neutral block. Then every family for a context C is also a family for the context obtained by adjoining the arms and the ports of B to the owner of C, with the same head, the same pinned input and the same token. If B is a token block the same holds provided C has at most one token, and the enlarged family carries the triple of B as its token when C has none and as its second token when C has one.*

Proof. Give B a parameter t that the family for C does not use. Every label of C is then zero in t and, by (N4), no label of B is, so the labels of the enlarged cell are nonzero, they are pairwise distinct across the two groups because they differ in the coordinate t, and they are pairwise distinct inside B again by (N4). By (N1) the owner keeps the output it had, and the only other outputs that change are those created inside B, so $P = O$ for the enlarged cell follows from $P = O$ for C together with (N2). Condition (N3) splits the new labels into pairs of opposite forms, which extends the signature partition by mirror pairs and leaves the token where it was. For a token block the same partition is extended by the mirror pairs together with the new triple, which sums to zero and spans a plane as the signature demands, and which is disjoint from the token of C because it lives in the fresh parameters. Finally the head chain, the pinned input, the token and the ports of C are untouched, so the free-head condition, the input condition, the token rank and each of the variant conditions of Section 4 hold for the enlarged family exactly as they held before. □

The two blocks we use most often are neutral for a reason one can write down.

Lemma 3.7. *Two arms of the same length ℓ form a neutral block, and so do two ports. In addition, two arms of lengths $a < b \leq 8$ of the same parity form a neutral block for every such pair except $\{1,3\}$, which is not neutral; one port together with an arm of even length $\ell \in \{4,6,8\}$ is neutral; one port together with two arms of opposite parity is neutral; and two ports together with an arm of odd length $\ell \in \{5,7\}$ are neutral. In addition, two arms of opposite parity $\{o,e\}$ form a token block for every pair with $o \in \{3,5\}$, $e \in \{2,4,6\}$ and for $\{1,6\}$ and $\{2,5\}$, although $\{1,2\}$ and $\{1,4\}$ do not.*

Proof. For two arms of length ℓ take $a_i = (-2)^i t$ on one and $a_i = -(-2)^i t$ on the other. The outputs of the first are $a_i + a_{i+1} = -a_i$ for $i < \ell$ together with a_ℓ, and those of the second are their negatives, so the $2\ell + 2$ outputs are exactly the $2\ell + 2$ labels $\pm(-2)^i t$, $0 \leq i \leq \ell$. Condition (N1) holds because the two first labels are t and $-t$, condition (N3) because the labels come in opposite pairs, and (N4) because the 2^i are distinct. Two ports labelled t and $-t$ satisfy the four conditions directly. For the remaining blocks, and for the assertion that $\{1,3\}$ admits no neutral labelling, the four conditions are a finite integer feasibility question once a bound on the c_i is fixed, and we settled it by an exhaustive search over integer labels with coefficients at most 8 in absolute value; the resulting labels are listed in Table 1, and each is re-verified from its labels alone, with no reference to how it was found. ☐

That $\{1,3\}$ is the one exception does not appear to be an accident of the search. It stays infeasible when the block is allowed three parameters instead of one and coefficients up to 14 in absolute value. We have not proved that no labelling whatsoever works, and we do not claim it; the pair is in any case the one our construction singles out everywhere else as well. A $\{1,3\}$ block can still be carried by a cell, as the catalogue shows; what fails is only the uniform statement that it may be adjoined to an arbitrary family.

Lemma 3.6 is what keeps the catalogue within reach. A context with a retained neutral unit, or with a pair of recruited ports, is almost always a narrower context with a block adjoined, and the wide contexts are exactly the ones a solver finds hard. Of the 135 contexts our searches had not reached, 128 are obtained in this way from a narrower context that already had a family, at no search cost at all, a further 20 input-in-token variants are obtained the same way, since by Lemma 3.6 the adjunction does not move the pinned input out of the token, and the last 8 contexts, all of them with an $\{o,e\}$ residual and no port, are obtained from a token block. Together with the searches this leaves no context of Lemma 3.8 without a family. The construction is not trusted: the families it produces are written in the format the solver emits and are re-verified by the independent checker of Section 8 together with everything else.

Lemma 3.8 (context completeness)**.** *Every owner receives exactly one context from a finite list, and every vertex of degree two lies in exactly one arm, head chain or neutral unit. Exactly 2,790 of the contexts of the list occur.*

The case analysis is given in Appendix A. Its regression, our executable scheduler, produces a context in the list for every owner of every tree with at most nine vertices (4,782,969 labelled trees) and of 169,000 uniform random labelled trees, 20,000 at each odd order $n \in \{11, 13, 15, 17, 19, 21, 31, 51\}$ and 3,000 at each of $n \in \{101, 201, 401\}$. No owner of any of them is left without a context and no context outside the list is produced. Which branch vertex the core is rooted at is a free choice of the decomposition, and it is one that matters: a root can be left with a residual that its arms and ports cannot cover and with no active child to spend on it. Our scheduler therefore tries the branch vertices in turn and keeps the first for which every owner receives a context. On 2,000 random trees of orders 11 to 51 the first branch vertex failed for 189 of them and every one of the 189 was covered at another branch vertex; we have not proved that a good branch vertex always exists, and we record that as the one place where our regression outruns our proof. In addition, the decision at a vertex depends only on its head-chain length, its arm lengths, its number of ports and the types of its active children. We therefore enumerated all 5,398,800 abstract local states with up to three arms of lengths at most eight (and larger arm sets of short arms), up to three ports, up to four active children of each of the possible types and head chains of length at most six. Each state was realised by a concrete tree fragment, and we recorded the context emitted at the vertex. The 2,790 distinct contexts obtained are all in the list, and they are exactly the contexts for which families are required in Section 4, whereas the remaining entries of the list are slack.

The cap on the number of active children is the one hypothesis of this enumeration that the decomposition does not force, since an owner of a large tree may carry arbitrarily many active children, and a tree on at most nine vertices carries none with three. We therefore ran the enumeration separately at the caps two, three and four. Raising the cap from two to three adds six contexts, all of them roots with a single arm, and 416 realizable ordered pairs, whereas raising it from three to four adds nothing at all: the 3,818,400 states enumerated at the cap four emit exactly the same contexts and exactly the same ordered pairs as the 1,204,800 states at the cap three. Thus the emitted set is stationary between three and four, which is what the analysis of Appendix A predicts, since the root rule and the two-input rule reduce any number of active children to one or two before a context is emitted. We note that a stationary pair of caps is evidence for, and not a proof of, stationarity at every cap; the proof is the case analysis itself.

Lemma 3.9 (charge)**.** *Let $q \leq D$ be the number of subdivided core edges. The number of owners is at most $2q$, the special support satisfies $|S| \leq 12D + 3$, and the values reserved for forced companions number at most $6q$.*

Proof. Every owner is incident with a subdivided core edge, and such an edge has two endpoints. A subdivided edge of length ℓ carries $\ell + 1$ labels, at most $D + q$ in total. Recruited ports contribute at most three per owner, absorptions at most two per owner, and the final bridge triple three. ☐

4 Local families

4.1 Symbolic families

A *family* for a context is a list of labels given as integer linear forms, divided by a common denominator L, in parameters y (the head), x or x_1, x_2 (inputs), token coordinates and internal parameters, such that

(E1) $P = O$ holds identically in the parameters;

(E2) the labels other than the head split into pairs of opposite forms and at most two *tokens*, triples of forms summing to zero (a second token is needed only for two-input cells whose non-head label count is even);

(E3) the label forms are pairwise distinct and nonzero;

(E4) the three coefficient vectors of each token span a plane, and a *carrier* of at most two parameters on which they already span a plane is recorded;

(E5) the head is an independent parameter (nonroot contexts);

(E6) the only label depending on an input alone is the antipode of that input, and no non-head label is a rational multiple of the head.

We call a family *forced-antipode* when some non-head label is identically minus the head. Such a label necessarily lies in a token $\{-x, a, x-a\}$ of the child, since its mirror is the head and the head is excluded from the signature, and a parent can then absorb the head x neither by a helper port nor by a mirror sibling, because the antipode of the head is already used inside the child. Two contexts of our catalogue carry only forced-antipode families, the (1; one port) bottom and the two-input cell with head chain of length two and residual $\{1\}$, and only the second of them occurs, the first being folded into its parent. Every other context has a family that is not forced-antipode, so a single child context is responsible for the whole of what follows.

A parent absorbs the head of a forced-antipode child through a token of its own that contains the shared edge, $\{x, b, -x-b\}$, so that the two tokens are negatives of each other once the child's carrier is matched, $a = -b$. A stationary parent with at least two ports does this with its port block. An owner uses an *input-in-token* family of its context, found by the same search with the shared edge required to be a token member (and with a second, free token when the number of non-head labels is even). A parent whose context carries no such family instead absorbs the child by joining its rows into its own cell, the joined child type A2 of Appendix A, and the joined cell then has a family that is not forced-antipode, so that it is an ordinary child of the next parent and the absorption stops after one step. Although the two rules look interchangeable, the second is the one that is always available, and the first is what keeps the cells small.

One parent context deserves a separate remark, since it shows that the joining rule cannot be dispensed with. Let v be an owner with head chain of length $\equiv 0$, one arm of length 1, no ports and two kept inputs x_1, x_2, write a_1, a_2 for its arm labels and h for its head, and read off the three outputs $h + a_1 + x_1 + x_2$, $a_1 + a_2$ and a_2. The leaf output is a_2, and $a_1 + a_2 = a_1$ would force $a_2 = 0$, so $a_1 + a_2 = h$ and therefore $h = -(x_1 + x_2)$, which leaves $a_1 = t$, $a_2 = -(x_1 + x_2) - t$ with t free. The four non-head labels admit no token, and pairing t with $-(x_1 + x_2) - t$, or x_1 with x_2, forces $h = 0$. Thus $t = -x_1$ or $t = -x_2$, and in either case the labels are $-x_1, -x_2, x_1, x_2$. Thus this owner always carries the antipodes of both of its inputs, it has no input-in-token family, and a forced-antipode child under it must be joined. In the joined cell the second child is pinned by one label whose output is that label itself, which is exactly the role of a port, so the joint is searched as the context with one more port and one active child. Property (E6) is what makes adjacent cells compatible without any case analysis, since a parent forces at most the antipode of its input and a child never carries the antipode of its own head, so that no two forms of adjacent cells coincide identically. Property (E6) has two clauses and both are needed for that argument, the first for the parent and the second for the child. We test them together and literally, over every family of every menu and, for a two-input row, against both of its inputs; a separate program does this, since our checker verifies (E1)–(E5) only. By that test, for 141 of the 2,789 contexts that occur and carry a family our catalogue has no family with (E6), and for these the finitely many adjacent pairs are checked individually by the gate of Section 8. We note that the gate does not rely on this count, since it enumerates every ordered pair of contexts and not only the pairs that (E6) leaves open.

4.2 The search and the checker

Families are found by our constraint model over the integer coefficients of all label forms: one Boolean per (output, label) pair encodes the permutation, each true Boolean imposes the equality of two coefficient vectors, the signature (mirror pairs, token, head, inputs, zero label) is encoded by further Booleans, and every coefficient vector is required to be nonzero and distinct from the others. The model is solved with CP-SAT [9]. A solution is a symbolic family, not a numerical labelling, and it is accepted only after an independent program reconstructs the topology from the row specification, recomputes the outputs, and re-verifies (E1)–(E5) with exact rational arithmetic and at three random integer points. Every family in the table passes, and in addition the six-edge mother and the twelve-label insertion are applied once to every arm and every head chain of every family and the result is re-verified. Table 2 summarises the catalogue.

Four exact impossibilities shape the alphabet. A single arm of length four with no ports has only one-parameter families (rays), so a residual even singleton retains one neutral unit. A head chain of length one or two with ports only has no $P=O$ family at all, and a subdivided parent edge therefore does not by itself make a vertex an owner. The residue-one letter on a direct parent edge has the unique family $(-x;\, -x-t,\, t;\, x)$, whose head is the antipode of its input and is therefore not a free parameter, and it is for this reason joined with its parent. With a subdivided parent edge the same letter has ordinary families with a free head, e.g. $(y;\, -y-x,\, -x,\, y+x;\, x)$ for $h=1$, and it is then an owner of its own. Finally, the two-input cell with head chain of length two and residual $\{1\}$ has no family without the antipode of its own head, over all denominators $L \leq 12$, all numbers of internal parameters up to three, all coefficient bounds up to 24 and all four token shapes, and this is what makes it the one forced-antipode context that occurs. The classification of the direct-edge two-input row given above is of a different kind, since it is proved by hand and does not depend on the search at all.

5 Sequential realization

Lemma 5.1 (realization)**.** *Let M be the least common multiple of the catalogue denominators and of 24. There is an absolute constant A such that, for every decomposed core with a family assigned to every owner as in Section 4, parameter values in $M\mathbb{Z} \cap [-A(D+1), A(D+1)]$ can be chosen bottom-up so that all special labels are distinct and nonzero and the special support is closed under negation up to at most one zero-sum triple whose negatives are unused.*

Our proof maintains the set of placed labels, the values reserved for the parent's forced companions, and at most two pending tokens whose carrier parameters are kept symbolic. A token cell with nothing pending keeps its carrier symbolic and every other parameter numeric. A token cell with a pending token, on the other hand, solves the two pairing equations for its carrier as affine functions of the live parameters and then fixes them in a box avoiding the recorded forbidden conditions, and all other cells choose their parameters one coordinate at a time. Because the only symbolic labels belong to the pending cells, their active children and their parents, at most $O(1)$ forms are alive and the number of forbidden conditions stays $O(|S|)$, and thus Lemma 3.9 gives boxes of side $O(D+1)$. The full argument is given in Appendix B, together with the explicit value $A = 8 \cdot 10^{12}$. This value follows from the catalogue constants (width at most 29, at most 7 parameters, coefficients at most 64 over denominators dividing 12, determinants at most 144 over the carriers we choose) and Lemma 3.9. Solving the pairing equations divides by the determinant of the carrier, which enlarges the lattice to $3{,}456\mathbb{Z}$ and the coefficients to 27,648 once, and it is these enlarged constants that A is built from. We have made no attempt to optimise it.

6 Dense completion and the proofs of Theorems 1.1 and 1.2

After Lemma 5.1 the unused labels form the set $X = \mathbb{Z}_n \setminus (\{0\} \cup S)$, which is closed under negation up to the pending triple T. The negatives $-T$ are placed in an odd ordinary block of size at least three, which exists by parity whenever a token is pending and no block has size one; write r for that block's prescribed size and put $X' = X \setminus (-T)$. Since r is odd and at least three, the remainder $r-3$ is zero or at least two, and the block is dropped from the list when it is zero. It remains to partition X' into zero-sum blocks of the resulting sizes, all at least two, with $|\mathbb{Z}_n \setminus X'| \le 2A(D+1)+4$.

Lemma 6.1 (linear dense completion)**.** *There are absolute constants $\eta > 0$ and n_1 such that for every odd $n \ge n_1$, every symmetric $X \subseteq \mathbb{Z}_n \setminus \{0\}$ with $|\mathbb{Z}_n \setminus X| \le \eta n$, and every list of block sizes $r_w \ge 2$ summing to $|X|$, the set X has a partition into zero-sum blocks of sizes r_w.*

Proof of Lemma 6.1. Write $n = 2K+1$. Since n is odd and X is symmetric with $0 \notin X$, the set X is a disjoint union of inverse pairs $\{j, -j\}$; let $J \subseteq [1, K]$ be the set of magnitudes occurring, so that $|X| = 2|J|$ and $|\mathbb{Z}_n \setminus X| = 1 + 2(K - |J|)$. In particular $|X|$ is even, so the number o of odd block sizes is even. Let m_2 be the number of blocks of size two and put $N = \sum_{r_w \ge 3} r_w$, so that $|X| = N + 2m_2$. We fix $\eta \le 1/20$ and argue by cases according to the size of N.

Case 1: $N \le n/100$. In this case we use no external input. A *twin* is a pair of magnitudes $a < b$ in J with $a+b \in J$; it yields the two zero-sum triples $\{a, b, -(a+b)\}$ and $\{-a, -b, a+b\}$, which are disjoint and whose union $\pm\{a, b, a+b\}$ is symmetric. We choose $o/2$ twins with pairwise disjoint magnitude sets, greedily. At the start of step s the forbidden magnitudes are those outside J, of which there are at most $\eta n/2$, together with the $3(s-1) \le 3o/2 \le N/2 \le n/200$ already used. Hence at most $(\eta + 1/100)K + 2$ magnitudes are forbidden. On the other hand the number of triples $\{a, b, a+b\} \subseteq [1, K]$ with $a < b$ is $\sum_{c=3}^{K} \lfloor (c-1)/2 \rfloor \ge K^2/5$ for K large, whereas a single magnitude lies in at most $2K + K/2 + 3$ of them. Since $(\eta + 1/100) \cdot (5/2) < 1/5$ for $\eta \le 1/20$, a permissible twin survives at every step once n is large, although the margin is not optimised. Now pair the odd blocks arbitrarily, give the two triples of the s-th twin to the two blocks of the s-th pair, and distribute the remaining magnitudes as inverse pairs, $p(r_w) = \lfloor r_w/2 \rfloor - [r_w \text{ odd}]$ of them to the block w. The count is exact, because $\sum_w p(r_w) = (|X| - 3o)/2 = |J| - 3o/2$ is precisely the number of magnitudes left over. Every block then has its prescribed size and sum zero.

Case 2: $N > n/100$. Here we run the argument of [7, Lemma 6.27] in $\mathbb{Z}_n$, which has no involutions, and with the missing pairs held out. Put $p = N/(2|J|)$, so $1/100 \le p \le 1$ because $2|J| = |X| \le n$. Enumerate all K inverse pairs of $\mathbb{Z}_n$ and let Y be a p-random set of indices; by [7, Lemma 6.26] applied to the partition of $\mathbb{Z}_n$ into these pairs together with $\{0\}$, the union of the selected pairs splits as $Q \sqcup R \sqcup S$ with Q, R disjoint $p/2$-random subsets of $\mathbb{Z}_n$ and $S \subseteq \{0\}$. Let Y_J be the set of selected indices whose magnitude lies in J; it is p-random on $|J|$ indices with integral mean $p|J| = |J| - m_2$, so its median equals its mean and, with probability at least one half, $|Y_J| \le |J| - m_2$, while Chernoff's bound gives $|Y_J| \ge |J| - m_2 - \sqrt{n}\log n$ with high probability. Fix an outcome in which both hold and in which Q and R satisfy the conclusions of [7, Lemmas 6.23 and 6.25]. Reserve m_2 of the $|J| - |Y_J|$ unselected available pairs for the blocks of size two, let J^* be the union of the at most $\sqrt{n}\log n$ remaining unselected available pairs, and put $Z = J^* \cup \bigcup_{i \in Y_J} \{i, -i\}$, a union of inverse pairs with $|Z| = 2|J| - 2m_2 = N$ and $\sum Z = 0$.

Reduce the multiset $\{r_w : r_w \ge 3\}$ to a multiset $M' \subseteq \{3, 4, 5\}$ as in the first paragraph of the proof of [7, Lemma 6.25], recombining at the end, and split $M' = M^Q \sqcup M^R$ by giving M^Q the floor and M^R the ceiling of half the multiplicity of each of $3, 4, 5$. The multiplicities need not be even, so the two parts need not have equal sums; however they differ in multiplicity by at most one in each of the three sizes, hence $|\sum M^Q - \sum M^R| \le 12$, and a bounded discrepancy is absorbed by the slack reserved below.

Write $\delta = (p/2)^{10^{10}}/\log(n)^{10^{24}}$ for the perturbation radius that [7, Lemma 6.25] allows at density $p/2$, and let $\varepsilon = \delta/24$, which lies in the range permitted by [7, Lemma 6.23] once n is large. Both $Q \setminus Z$ and $R \setminus Z$ are contained in $\{0\} \cup (\mathbb{Z}_n \setminus X)$ and hence have size at most $|\mathbb{Z}_n \setminus X|$, and $|\sum M^Q - pn/2| \le |\mathbb{Z}_n \setminus X| + O(\sqrt{n}\log n)$. Assume $|\mathbb{Z}_n \setminus X| \le \delta n/8$, which is the hypothesis of the lemma with $\eta = \delta/8$. Then [7, Lemma 6.23], applied with $R = Q$, Z as above, $m = \sum M^Q$ and $g = 0$, returns $Q' \subseteq Z$ with $|Q'| = \sum M^Q$, $\sum Q' = 0$ and

$|Q'\triangle Q| \le 6\varepsilon n = \delta n/4$. Put $R' = Z \setminus Q'$; then $|R'| = \sum M^R$, and $\sum R' = \sum Z - \sum Q' = 0$, and

$$|R'\triangle R| \le 6\varepsilon n + |\mathbb{Z}_n \setminus X| + |J^*| \le \tfrac{\delta n}{4} + \tfrac{\delta n}{8} + O(\sqrt{n}\log n) \le \delta n$$

once n is large, and likewise $|Q'\triangle Q| \le \delta n/4 \le \delta n$. Thus both perturbations stay strictly inside the allowance, which is the point of taking ε a twenty-fourth of δ rather than a sixth. Therefore [7, Lemma 6.25] applies to Q' with the reference set Q and the multiset M^Q, and to R' with R and M^R, and returns zero-sum partitions of Q' and of R' of the prescribed sizes. Together with the m_2 reserved inverse pairs these fill every block, since $Z \sqcup (\text{reserved pairs}) = X$. When $m_2 = 0$ one has $p = 1$, Y_J is all of J and J^* is empty, and the argument degenerates to a direct application of [7, Lemma 6.25] to X. Thus X has the required partition whenever $|\mathbb{Z}_n \setminus X| \le \delta n/8$, and δ depends only on $p \ge 1/100$ and on n through the logarithm. $\square$

Two readings of [7] enter through the parameter ε of the last proof, and they give the two thresholds of Section 1. The printed perturbation radius of their Theorems 4.3–4.7, and hence of their Lemmas 6.23 and 6.25, is $p^A n/(\log n)^A$; since the proof above uses those lemmas only with $p \ge 1/200$, this gives Lemma 6.1 with ηn replaced by $n/(\log n)^A$, and hence Theorem 1.1. Their Section 7, under *Bounds in the main theorem*, records that $\log n$ may be replaced by $\log |G'|$ throughout the proof, the only source of logarithms being their short commutator-product theorem, and for an abelian group the commutator subgroup is trivial and the commutator product is empty; the authors record the consequence $g(n) = \Omega(n)$ for cyclic groups in their Problem 7.4. One should read their commutator length as bounded by $\max(1, \log_4 |G'|)$ rather than divide by $\log 1$. Their Lemmas 6.23 and 6.25 are deduced from Theorems 4.4–4.6 with no further logarithmic loss, the auxiliary error terms being of order $\sqrt{n}\log n$, so at every fixed reservoir density this reading leaves a positive constant perturbation radius and gives Lemma 6.1 as stated, which is the hypothesis of Theorem 1.2. This improvement is described by the authors as requiring essentially no modification of their proof, although neither it nor its consequence for their Section 6.3 is a printed theorem. The constant η is in either case inherited from their argument and is not effective.

Proof of Theorems 1.1 and 1.2. Root T at a branch vertex, decompose by Section 3, assign families by Section 4, realize by Lemma 5.1 with $|S| \le 12D + 3$ and magnitudes at most $A(D+1)$, and place the pending triple. Under the hypothesis of Theorem 1.2, Lemma 6.1 completes the labelling as soon as $2A(D+1)+1 \le \eta n$; every stationary vertex then has a zero-sum block, every special vertex has the prescribed sum, and Lemma 2.1 gives the bijection, so $C \ge 3A/\eta$ and $n_0 \ge n_1$ prove Theorem 1.2. With the printed perturbation radius of [7] in place of the linear one the same argument applies when $2A(D+1) + 1 \le n/(\log n)^{A'}$, which proves Theorem 1.1; its final assertion follows by taking $D \le k$ and n large. $\square$

7 An elementary route to the linear threshold

Theorem 1.2 rests on the abelian reading of [7] through Lemma 6.1. Here we reduce Lemma 6.1 to a single statement about permutations of an interval, prove that statement at the two ends of its range and for a family of composite orders, and verify it exactly up to order 30. A proof in general would give Theorem 1.2 unconditionally, with an explicit C.

7.1 From the residual set to Heffter's difference problem

After Lemma 5.1 the special support is $\pm I$ with $I \subseteq [1, M]$ and $M = A(D+1)$, and the residual set is $X = \pm([1,K] \setminus I)$ in $\mathbb{Z}_n$, $n = 2K+1$. In the branch of Lemma 6.1 that needs the completion theorem, X has to be partitioned into signed zero-sum triples $\{\pm a, \pm b, \pm c\}$. Writing a triple by its three magnitudes $a < b < c$ in $[1,K]$, the two possibilities are $a + b = c$

and $a+b+c=n$; the first is a pair $\{b,c\}$ at distance a and the second is the pair $\{b,-c\}$ at cyclic distance a in $\mathbb{Z}_n$. For $I=\emptyset$ this is Heffter's first difference problem when $n\equiv 1 \pmod 6$, solved for every n by Peltesohn [8], and the general case asks for the same partition of $[1,K]\setminus I$, that is, for Heffter's problem with the small differences I forbidden. The hardest case in our experiments is $I=[1,M]$, and it is the one we analyse.

7.2 Folded sum permutations

Let $K-M=3t$. In every solution our searches found, the t smallest elements $M+1,\dots,M+t$ can be taken as the distances a, and the remaining $2t$ elements split into a lower half $M+t+1,\dots,M+2t$ and an upper half $M+2t+1,\dots,K$ with each pair $\{b,c\}$ taking b from the lower and c from the upper half. Writing $a=M+1+i$, $b=M+t+1+j$, $c=M+2t+1+k$ with $i,j,k\in[0,t-1]$ and $s=M+1-t$, the two triple types become

$$c-b=a \iff k=i+j+s, \qquad b+c=n-a \iff k=(2t-1)-(i+j+s).$$

So a solution is a permutation π of $[0,t-1]$, $j=\pi(i)$, such that the numbers $u_i=i+\pi(i)+s$ lie in $[0,2t-1]$ and their folds $\min(u_i,\,2t-1-u_i)$ form a permutation of $[0,t-1]$. We call such a π a *folded sum permutation* of order t and shift s, and write $\mathrm{FSP}(t,s)$. Equivalently, with $\tau_i=i+\pi(i)$, the τ_i are distinct, lie in the window $W=[-s,2t-1-s]$, and no two of them sum to the reflection constant $c=2t-1-2s$ of W; that is, the sum set of π is a transversal of the reflection of W. Since $\sum_i \tau_i=t(t-1)$ for every permutation, the mean of the transversal exceeds the centre of W by $s-\frac{1}{2}$.

The range $3M+1\le K\le 7M+3$ of Lemma 6.1 corresponds to $(1-t)/2\le s\le (t+1)/2$, and beyond $K\ge 7M+3$ Simpson's theorem on Langford sequences applies with no folding at all.

Conjecture 7.1. *For every $t\ge 5$ and every integer s with $(1-t)/2\le s\le (t+1)/2$ there is a folded sum permutation of order t and shift s.*

The window is sharp: outside it no $\mathrm{FSP}(t,s)$ exists, because the s largest values of W (for $s\ge 1$) have no reflection partner in the range of the τ_i and must all be attained, which is impossible once $s>(t+1)/2$. Our constraint programs verify the conjecture exactly, for every $3\le t\le 30$ except $t=4$, where $s=-1$ and $s=2$ fail; unlike Simpson's theorem there are no congruence exceptions inside the window, the folding removes them.

7.3 What is proved

Proposition 7.2. *Conjecture 7.1 holds in the following cases.*

1. *$s\in\{0,1\}$: the identity is an $\mathrm{FSP}(t,s)$.*
2. *t odd and $s\in\{(1-t)/2,(t+1)/2\}$: the rotation $\pi(i)=i+(t-1)/2 \bmod t$ is an $\mathrm{FSP}(t,s)$ at both endpoints.*
3. *(Composition.) If a is odd and σ is an $\mathrm{FSP}(b,s')$, then*

$$\pi(au+w)=a\,\sigma(u)+\left(\left(w+\tfrac{a-1}{2}\right) \bmod a\right), \qquad u\in[0,b-1],\ w\in[0,a-1],$$

is an $\mathrm{FSP}(ab,s)$ with $s=a(s'-1)+(a+1)/2$.

Proof. (1) The sums $2i$ are distinct and even, the constant c is odd, and $0\le 2i\le 2t-2\le 2t-1-s$ exactly when $s\le 1$; the lower bound $2i\ge -s$ needs $s\ge 0$.

(2) For the rotation the sums $i+\pi(i)$ are $2i+(t-1)/2$ for $i<(t+1)/2$ and $2i-(t+1)/2$ for $i\ge (t+1)/2$, which together form the interval $[(t-1)/2,(3t-3)/2]$. At $s=(1-t)/2$ this

interval is the lower half of W, at $s = (t+1)/2$ it is the upper half, and a half of W contains no reflected pair.

(3) On each block the inner rotation has sums $w + \rho(w)$ forming the interval $I = [(a-1)/2, (3a-3)/2]$, which is carried to itself by $x \mapsto 2a-2-x$. The sums of π are $a(u+\sigma(u)) + I$. Put $s = (a+1)/2 + aq$. The condition $a(u+\sigma(u)) + I \subseteq [-s, 2ab-1-s]$ is $u + \sigma(u) \in [-(q+1), 2b-q-2]$, the window of $\mathrm{FSP}(b, q+1)$; and the reflection constant $c = 2ab-1-2s = a(2b-2q-1)-2$ sends $a\,y + x$ to $a(2b-2q-3-y) + (2a-2-x)$, so it reflects I onto itself and the block index y by the constant $2b-1-2(q+1)$ of $\mathrm{FSP}(b, q+1)$. Therefore the sum set of π is a transversal exactly when the sum set of σ is, and $s' = q+1$. $\square$

Case (2) has two readings: at the lower endpoint it is a hooked Langford sequence of defect $(t+1)/2$, at the upper endpoint the columns $(i, \pi(i), c-i-\pi(i))$ form a $3 \times t$ magic rectangle. Case (3) covers the residues $s \equiv (a+1)/2 \pmod a$ for every odd divisor a of t; it does not reach prime t, and we have not found an additive counterpart.

7.4 Consequences

If Conjecture 7.1 holds then the clean tail $[M+1, K]$ has a maximum packing into difference triples for every $K \geq 3M+1$ (the cases $K - M \not\equiv 0 \pmod 3$ reduce to it by omitting the one or two smallest elements, and the finitely many cases $K \leq 3M+3$ are magic rectangles with one or two empty cells, verified for $M \leq 30$). Together with Simpson's theorem on Langford sequences, which covers $K \geq 7M+3$ with no folding, this gives Lemma 6.1 with $\eta = 1/(14A + O(1))$, hence Theorem 1.2 unconditionally, with an explicit C and without any use of [7]. The punctured cases $I \subsetneq [1, M]$ are the same object with the extra small elements as additional distances, and were found feasible for every random I we tried; we have not reduced them to Conjecture 7.1 in general.

8 Computations and certificates

Three programs enter the proof, and we describe them in the order in which their results are used.

The symbolic search. For a context the search builds the label forms of Section 4 with integer coefficients bounded by 24 over a common denominator $L \leq 12$ (the families built by the two identities of Section 3 are not subject to this cap, and the largest coefficient anywhere in the catalogue, 64, is attained by the equal-pair blocks; it is this catalogue-wide value that enters Lemma 5.1), encodes the $P = O$ identity as a permutation of the output multiset onto the label multiset (one Boolean per pair, each true Boolean equating two coefficient vectors), encodes the signature (E2), the distinctness (E3), the token rank condition (E4) and the head freedom (E5) as further Boolean and linear constraints, and asks CP-SAT [9] for a feasible assignment. The same model with the cleanliness conditions (E6) is tried first, and the alternatives (token of rank three, token of rank two, no dedicated token parameters, two tokens) are tried in that order. Every feasible assignment is stored, and the menu of a context lists all of them in order of preference, families containing the antipode of their own head last. For the contexts with inputs the model is run a second time with the shared edge required to be a token member and the antipodes of the inputs forbidden (and a third time, for two-input cells, with both inputs required to be token members). The resulting *variant* families are kept in separate tables and are the ones used when a child is forced-antipode. Each context is tried under nine settings of (denominator, number of internal parameters, coefficient bound), namely $(1,1,3)$, $(2,1,3)$, $(3,1,3)$, $(1,2,4)$, $(2,2,4)$, $(3,2,4)$, $(4,2,4)$, $(6,2,6)$ and $(12,2,12)$, with the head multiplier free, and with a time limit between 150 and 900 seconds per context and setting on a 128-core workstation. In all, 14,041 contexts with widths up to 29 labels were attempted, in 30,479 runs of the solver, and 5,446 of them received at least one family. The search is exact in the sense that an infeasible model is a proof of non-existence within the coefficient and denominator bounds, and it is in this sense that the impossibilities of Section 4 are stated.

The independent checker. The checker reads only the JSON record of a family, rebuilds the cell topology from the row specification (head chain, arms, ports, active and input children, root row), recomputes every output as an exact rational combination of the parameters, and verifies (E1) by matching the two multisets of coefficient vectors, (E2)–(E5) directly, and the numerical identity at three random integer points, and it shares no code with the search. All 12,742 families of the catalogue and all 2,587 variant families pass, as do the families obtained from them by one six-edge insertion on every arm and by one twelve-label insertion on every head chain. The command

```
python3 scripts/check_alphabet_families.py data/alphabet_families.json
```

re-verifies the whole catalogue in about two minutes on a laptop and prints one line per context.

The gate and the closure check. Two adjacent cells can fail to be compatible in only one way. A label of the parent that depends on its input alone is qx for a rational q, a label of the child that depends on its own head alone is $q'y$, and if the two ratios agree once the shared edge is identified then the two labels are equal as forms and no choice of parameters separates them. Our gate enumerates every ordered pair of contexts, tries every pair of families from the two menus, and reports the context pairs for which no combination avoids such a coincidence. Over the catalogue it reports 1,973 ordered pairs. However, most ordered pairs of contexts cannot occur in a tree at all, and restricted to the 11,927 ordered pairs that the enumeration of Lemma 3.8 records it reports 71. All but one of these have the forced-antipode child of Section 4 as their child and are therefore absorbed by the rule of that section and not by the gate. The exception appeared only when the enumeration was run at three active children, and it is the pair whose parent is the root context with a lone arm of residue two and one active child, and whose child is the bottom cell with arms of lengths one and two. Here the parent's forced ratios are -1, $-\frac{1}{2}$ and $\frac{1}{2}$ in every family of its menu, and the child carries the antipode of its own head in every family of its menu, so the two collide at every head multiplier the search produces. However, the gate is a sufficient condition and not a necessary one: we join the child into the parent and solve the two label sets together as a single cell of width ten, for which the catalogue carries a family, and an identity between labels of one cell cannot arise. We note that the joining stops there, since the parent is the root and the child is a bottom cell, so the enlarged cell has neither a parent of its own nor an active child, and this is the same device as the joint of the absorption rule. A second program checks the conditions that the absorption rule needs, namely that every context that occurs has a family, that every context with a free active child has an input-in-token family or a joint that is not forced-antipode, that the folded letter has such a joint, and that every two-input context either has a family with both inputs in tokens or has a port joint carrying the second pinned child in a token. We note that our scheduler deliberately stops one step short of the absorption rule, since it emits the forced-antipode cell as an owner of its own and does not join it into its parent, because which of the two routes applies depends on the certificate table and not on the tree. The context list therefore carries the joint of every free-child context and the port joint of every two-input context, and the second program is what verifies that the rule closes over them. These conditions are a finite list and the program reports them one by one, so that the state of the verification is visible at any time. All of them hold: every one of the 2,790 contexts that occur has a family, all 544 free-child contexts absorb a forced-antipode child, the folded letter and all 40 two-input contexts do the same, and the gate leaves nothing outside the absorption rule and the single join above.

The scheduler and the constructor. Our scheduler implements the rules of Appendix A and returns, for a rooted tree, the list of owners with their contexts. It is the regression for Lemma 3.8 and was run on all labelled trees with at most nine vertices and on 169,000 random trees of odd orders between 11 and 401. It also chooses the branch vertex at which the core is rooted, as described after Lemma 3.8. Our constructor implements Lemma 5.1 on top of the scheduler and the catalogue, produces an integer labelling, reduces it modulo n, and checks the bijection of vertex sums directly. Each run writes a certificate listing the owners, the chosen families, the parameter values and the final labelling, so that a labelling can be re-verified without the constructor. The constructor has two modes. In its default mode it follows the decomposition but completes the ordinary blocks by a direct search, and in the sparse regime ($n \in \{51, 101, 201\}$ with $D \approx n/25$) it succeeded on 116 of 120 random trees. In its *proof-faithful* mode it follows Lemma 5.1 step by step (numeric coordinates, live carriers, deferred pairing, the reservation of forced companions) and reports every departure from the lemma as a named gap. On the same 120 trees it succeeded on 98. Of the twenty-two remaining runs, fourteen report a modular collision between two special labels, or a pending triple whose negatives are already used modulo n, seven report a failure of the completion step, and one reports that no odd block was available for the singleton bridge. We note that these runs use the dense completion only through a direct matching of the leftover labels, since at these orders the asymptotic Lemma 6.1 does not apply, and their purpose is to confirm the bookkeeping of Lemma 5.1 and not the theorem itself. A modular collision between two special labels is exactly the event that the hypothesis $n > C(D + 1)$ excludes, and at $n = 201$ with $D \approx 8$ the hypothesis is very far from holding. The runs also give the bookkeeping quantities of Appendix B directly: at most two parameters were ever live at once, at most sixteen labels were symbolic, at most two tokens were pending, and the largest label magnitude was 692, about $76(D + 1)$ at these orders, against the bound $A = 8 \cdot 10^{12}$ that we prove.

9 Discussion

Previous work on edge-graceful trees is of two kinds. Explicit labellings are known for several families, for example paths, stars, spiders and brooms (see [2] for the list; the regular spiders of [1] already carry unboundedly many vertices of degree two, although in a single prescribed shape), and the zero-sum partition theorem of [3] covers every odd-order tree with at most one vertex of degree two. Our earlier work [6] extended that theorem to two such vertices with explicit (and effective) constructions. Our result lies between the two: it covers all trees in which the vertices of degree two are a small fraction of the order, with no restriction on where they sit, at the price of an ineffective constant.

The constant C is not effective because Lemma 6.1 is not, whereas the constant A of Lemma 5.1 is explicit ($A = 8 \cdot 10^{12}$). An effective version of the zero-sum matching with linear deficiency would therefore make Theorem 1.2 effective without any change to the construction, and Section 7 reduces this to Conjecture 7.1. However, the degree-two vertices remain a bottleneck in a stronger sense: the special support has size linear in D because every degree-two vertex is non-stationary, so our argument cannot reach trees in which these vertices have positive density. In particular it says nothing about caterpillars with many subdivided edges or about spiders with long legs, for which the explicit constructions in the literature remain the only tool. Whether the density threshold can be raised, e.g. to $D = o(n)$ or to $D \leq cn$ for a small absolute c, seems to us the natural next question. The local families are indifferent to the density, so the obstacle is entirely in the completion step, where the leftover set would no longer be nearly the whole group.

Two further remarks concern the computer-assisted part. First, our catalogue is finite by design and every family in it is a rational identity in a few parameters, and an independent re-verification is for this reason possible at all. The search is only a way of finding the identities, and the proof does not depend on the search having been exhaustive except in the four impossibility statements of Section 4, each of them a finite enumeration, and one of the four structural facts we use, the classification of the direct-edge two-input row, is proved by hand. Second, although our scheduler's regression covers all small trees, it is evidence for, not a proof of, Lemma 3.8. The proof is the case analysis of Appendix A, and the scheduler is our way of keeping that analysis honest.

Use of AI. Anthropic's Claude, OpenAI's ChatGPT and Google's Gemini were used as coding and analysis assistants throughout this project. They implemented and executed the processing, reasoning, and statistical code, prepared documentation, surveyed literature, and assisted in preparing manuscript text.

10 Data and code availability

The search (`symbolic_free_token_cpsat.py`), the batch driver, the independent checker (`check_alphabet_families.py`), the scheduler (`context_scheduler.py`), the abstract enumeration (`enumerate_emittable_contexts.py`), the gate (`check_lookahead_gate.py`), the closure check (`closure_check.py`), the literal test of (E4) and (E6) (`audit_e4_e6.py`, `e6_coverage.py`), the non-constancy check of the carrier elimination (`carrier_nonconstancy.py`), the constructor, the context list, the certificate catalogue (`alphabet_families.json`) with its variant tables, and the certificates of the runs reported in Section 8 are available at https://github.com/lsmeng/sparse-edge-graceful and archived at https://doi.org/10.5281/zenodo.22287387 (concept DOI, resolving to the latest version).

A Context completeness

We prove Lemma 3.8 by describing the assignment step by step. The executable scheduler follows the same steps and is our regression for the case analysis. Process the core bottom-up and let v be a nonroot core vertex with parent core edge of length h, arm multiset A, k active children (children which are owners) and q ports. Since v has degree at least three in T we have $|A| + k + q \geq 2$.

Step 1 (arms). Lemma 3.1 partitions A into neutral units and a residual R of one of the six types. The units are $P = O$ with zero head sum and with support closed under negation, so they do not enter the cell of v. However, when a unit is needed at v itself (Steps 3 and 4) one of them is retained and its arms are added to the residual.

Step 2 (active children). Cancelling a set of $j \geq 2$ active children means prescribing their heads to sum to zero: for j even we use $j/2$ mirror pairs, and for j odd one free zero-sum triple together with $(j-3)/2$ pairs. These heads are free parameters of the children's families by (E5), with one exception, and a child of the joined types, whose head is forced, is paired with a sibling of free head, the sibling taking the negative of the forced head. The exception is a pair of two children that both have forced heads, which our scheduler does produce. It is also arrangeable, one level further down: a child whose head is forced in this way has context $(0, \{1\}, 0, \text{free})$, its own active child is of free type, and the head of that grandchild is a free parameter by (E5), so setting the grandchild's head makes the two forced heads negatives of one another. We note that this uses (E5) at the grandchild and not at the child, and that it does not recurse further, since the grandchild is of free type and is therefore not one of the rigid rows.

Whether all of the active children can be cancelled depends on what Steps 3 and 4 can then do with the residual. If cancelling all of them leaves v covered, we cancel all of them, and v has at most one active child left, whose head is the input x. Otherwise v must spend active children on the residual, and the rule is by the parity of their number k. For k even we keep two and cancel $k-2$, so that v is a *two-input* owner with inputs x_1, x_2; for k odd we keep one and cancel $k-1$. In both cases the number cancelled is at least two whenever it is positive, so the cancellation above applies. At the root with a lone residual arm there is one deviation: $k = 2$ keeps both, because keeping one would leave a single child to cancel, and every $k \geq 3$ keeps one.

The configurations in which the residual cannot be covered without active children, so that this rule is used, are the following, and they are the exact condition our scheduler tests. Off the root, v has no ports, its residual is a single arm, and no neutral unit was retained; then the residues modulo six that need an active child are $1, 2, 4, 6$ when the head chain has length 0 modulo six and all of $1, \dots, 6$ otherwise. At the root with a lone residual arm of residue r and q ports, the recruited port count of Step 5 has no feasible value exactly when $q = 0$ and $r \in \{2, 4\}$, or $q = 1$ and $r \in \{1, 3, 5\}$; for $q \geq 2$ there is always one. At the root there is one further case, the opposite-parity residual $\{1, 2\}$ with no ports. We note that the earlier draft of this step named only the portless residuals $\{1\}$ and $\{2\}$ and only $k = 2$; that was too narrow, and a referee exhibited a tree of order 19 on which our scheduler keeps two active children at a residual of residue four. With a subdivided parent edge such cells have ordinary families, e.g. $(y, -y-x_1-x_2;\ -x_1-x_2,\ y+x_1+x_2;\ x_1, x_2)$ for $h = 1$ and $R = \{1\}$ with token $\{x_1, x_2, -x_1-x_2\}$, some of them with two tokens. With a direct parent edge the exact search shows that no family with a free head exists, and v uses instead one of the two *rigid* rows $(-x_1-x_2;\ -x_2,\ -x_1;\ x_1, x_2)$ for $R = \{1\}$ and $(-\frac{3}{2}(x_1+x_2);\ -\frac{1}{2}(x_1+x_2),\ -(x_1+x_2),\ \frac{1}{2}(x_1+x_2);\ x_1, x_2)$ for $R = \{2\}$, whose head is forced by the inputs and which are treated like the residue-two ray in Appendix B. In our random-tree regression this pattern occurs at roughly one to two percent of the vertices.

Step 3 (empty residual). Suppose $R = \emptyset$. If no active child remains, v is stationary and its ordinary block consists of its q ports, which requires $q \neq 1$. If $q = 1$ there is a neutral unit or a cancelled group because $|A| + k + q \geq 2$, and one neutral pair is undone, so that R becomes that pair (a same-parity residual, handled in Step 4). If one active child of free type remains and $q = 1$ or $q \geq 3$, a helper port labelled $-x$ absorbs it. If $q = 2$, two ports labelled a and $-x-a$ absorb it with a token of rank two. If $q = 0$ then $|A| \geq 1$, so a neutral pair is undone and v goes to Step 4 with the pair as residual and the active child. A remaining child of a joined type is never absorbed by a helper (for the residue-one letter the helper would be $+x$, which is the head of the grandchild). Instead v receives the context $(h, \emptyset, p, c)$ with c the joined type.

Step 3a (forced-antipode children). The exact census shows exactly two contexts whose every family contains the antipode of its own head, the (1; one port) bottom and the two-input cell with head chain of length two and residual $\{1\}$. The first is folded into its parent at Step 3 in every case, as the child type L11, so it never becomes an owner, and the second is therefore the only forced-antipode child that occurs. Such a child c is absorbed as follows. If the parent is stationary with at least two ports, the port block containing the head of c absorbs it. If the parent is an owner whose context has an input-in-token family, that family absorbs it. Alternatively, and this is what happens for the contexts whose input-in-token search is infeasible, c is joined into the parent's cell as the child type A2, and the joint family is re-verified by the checker as a whole. The joined cell has a family that is not forced-antipode, so it is an ordinary child of the next parent and the chain stops after one step.

A two-input owner can have two forced-antipode children, and it then joins one of them and keeps the other pinned. In the joint the pinned child contributes one label whose output is that label itself, which is the role of a port, so the joint is searched as the same context with one more port and child type A2, and its family is required to carry the port in a token and to have no label identically minus the port. This is why we always join, and do not absorb, when both are possible: a cell that is joined into its parent is then a two-input cell with no forced-antipode children of its own, so the two grandchildren that the joint pins are ordinary, and no cell ever needs more than one token for an absorption.

The residue-one letter needs a word of its own, although it is not forced-antipode. With a direct parent edge and no port its own context has no family with a free head, so it is folded into its parent as the child type L1. With a subdivided parent edge, or with a port, it is an owner of its own. However, when its active child is forced-antipode the folding is not what we do: the forced-antipode child is joined into the letter instead, which gives the context $(h, \{1\}, p, \mathrm{A2})$, and that context has a family that is not forced-antipode, so the letter is an ordinary owner and its parent sees an ordinary free child. Therefore an L1 joint never carries a forced-antipode grandchild. (The opposite-parity bottom $(1, 2)$ was joined in an earlier version of the alphabet, although it has a family without the antipode of its head and is now an ordinary owner.)

Step 4 (nonempty residual). v is an owner with context $(h \bmod 6, R', p, c)$ where R' is R after the following repairs: a portless residual $\{1\}$, $\{2\}$, $\{4\}$, $\{6\}$ or $\{8\}$ with no active child retains one neutral unit, and a lone odd arm with a head chain likewise retains one unit (its portless cell has only rays). One unit exists in each case because v has at least two children. The number of recruited ports is $p = q$ when $q \leq 2$, otherwise $p \in \{1, 3\}$ with $q - p \neq 1$, so that the unrecruited ports form a block of size zero or at least two. Every tuple produced in this way is in the list, itself generated by the same rules.

Step 5 (root). The root has no head and its context is $(\mathrm{root}, R', p, c)$ with the same conventions, the root family containing the zero label. Exact search shows that a lone residual arm at the root needs two recruited ports, or none for odd residue and one for even residue. When the available ordinary children do not permit this, one neutral pair is retained, and when the root has a lone arm, no ports, no unit and $k \geq 2$ active children the exact root table decides between cancelling all active heads and keeping one or two active children. A root whose only children are one residual arm and one active child cannot occur, because branch vertices have three children.

Every owner is assigned exactly one context, every vertex of degree two lies in exactly one arm, head chain or unit, and therefore the cells and units are edge-disjoint and cover all subdivided edges. □

The steps above are also the specification of our scheduler, and the abstract enumeration described in Section 3 is our check that the list of contexts generated from these steps is complete: the residual types, the port counts and the child types are finite, the head-chain and arm lengths enter only through their residues modulo six (with the raw lengths 1 and 2 distinguished), and every combination is exercised by at least one fragment.

B Proof of the realization lemma

Families and the choice rule

Every family carries the properties (E1)–(E5) of Section 4, which our checker verifies. Property (E6) holds for a family of every context that occurs except the 141 of Section 4, and for those the gate of Section 8 checks the finitely many adjacent pairs individually; the choice rule below is what turns this into the statement the realization uses. For a token family let $K(F)$ denote its carrier, a set of at most two parameters on which the three token forms already span a plane. Property (E4) asserts that some such pair exists but does not name it, so the carrier is ours to choose, and the choice matters for the constants below. On a fixed pair of columns the three 2×2 minors of the token coincide up to sign, because the three forms sum to zero, so each candidate carrier has one well-defined $|\det|$. We prefer, in this order, a carrier with $|\det| = 1$, then one that does not contain the head, then one of least $|\det|$. The first preference is what Remark B.1 turns on, since a unimodular carrier needs no division at all; the second matters because the head is the value a cell passes to its parent. Over the catalogue this minimum is 1 for 69% of the tokens and never exceeds 144, and the largest value of $|r_\ell \operatorname{adj}(M)|$ over all labels at the chosen carrier, where M is the 2×2 carrier block and r_ℓ the carrier row of a label ℓ, is 216. A family is *input-carried* when $K(F)$ contains an input. For every owner we fix a family before any label is chosen, preferring families with (E6). When the context of a parent has no family with (E6), the finitely many pairs (parent family, child family) are checked for identical forms directly. With this choice no non-input label of a parent is identically equal, as a form in the shared head, to a label of its child: the parent's only input-only label is the antipode of its input, and the child has no label equal to the antipode of its own head.

State

Owners are processed in post-order. The state consists of the set Placed of numeric special labels, the set of *reserved* values (antipodes of heads that a parent will place later), and the set Open $\subseteq$ Placed of placed labels whose antipode is not yet placed and is designated (the head of an active child, the three labels of a pending token). It also carries at most two *pending tokens* T_A, which then belong to one and the same cell A, together with a *live set* Λ of at most four symbolic parameters (the carriers of the pending cell A, and the head of A's active child when a carrier contains the input), and the *symbolic labels* (labels that are affine forms in Λ: they belong to A, to A's active child and to A's parent, hence at most $3W$ of them, where W is the largest family width). Finally it carries the set Φ of recorded forbidden conditions, each a proper affine condition on Λ.

A numeric value v is *admissible* for a label if $v \neq 0$, v is not placed or reserved, and $-v$ is not placed or reserved, unless $-v$ is open and v is its designated antipode (the forced $-x$ of an absorbing family, a helper, or a coordinate of $-T$ when a token is cancelled).

The invariant is: (a) numeric labels are pairwise distinct and nonzero, and no two are antipodal except designated pairs; (b) at most two tokens are pending, and both belong to the same cell, so the symbolic labels are still confined to that cell, its active child and its parent; (c) every symbolic label is a non-constant affine form in Λ, and no two symbolic labels are identical or antipodal as forms; (d) Φ contains every condition under which a symbolic label would collide with a placed value or with another symbolic label; (e) $|\,\text{Placed}\,| + |\text{Reserved}| \leq |S| + \#\text{cells}$ and $|\Phi| \leq 3W(2|\,\text{Placed}\,| + 3W)$.

Processing an owner

Let v be the next owner with family F. Its input is the head label of the remaining active child, numeric unless that child kept its head symbolic, which happens exactly when F is input-carried (the active child of an input-carried owner always keeps its head symbolic and chooses its other parameters numerically) or when the child is the pending cell with its head in Λ.

No token. Choose the head and the internal parameters numerically, one coordinate at a time, in $M\mathbb{Z} \cap [-B, B]$, in a fixed order of the parameters. Property (E3) says that each label form is nonzero and that no two of them are equal, so each label form, and each difference of two label forms, has a well-defined *last* nonzero coordinate in that order. We charge every admissibility requirement and every reservation to the step at which the last nonzero coordinate of the form concerned is chosen; at that step the form is affine in the coordinate being chosen with nonzero slope, whereas at every later step it is already numeric. Each requirement therefore excludes at most one value, at exactly one step, and the reservation is $\{qy : q \in \Phi(F_{\text{parent}})\}$. We note that (E3) alone would not justify this coordinate by coordinate, since a label may well have zero coefficient in several parameters; it is the charging to the last nonzero coordinate that makes the count valid. The bound $B = M\big(2W(|\,\text{Placed}\,|+|\text{Reserved}|+1)+W^2+1\big)$ therefore suffices. If the input is symbolic, the labels depending on it are symbolic forms, non-constant by (E3) and the choice rule, and every comparison with a numeric label or another symbolic form is recorded in Φ.

Token, nothing pending. Keep the carrier symbolic, $\Lambda := K(F)$ (together with the child's head when the carrier contains the input), choose all other parameters numerically as above, and declare the token pending. If the head is in Λ the parent of v receives a symbolic input. Its own head is numeric by (E5), so no further propagation occurs. The rigid rows are the exception to (E5): the residue-two ray and the two direct-edge two-input rows of Appendix A have heads that are fixed forms in their inputs, so their parents receive a symbolic input whenever those inputs are symbolic. A chain of residue-two rays propagates a symbolic head for at most three levels before the four-layer reset family restores a free head, and a two-input row is processed only after both of its children, so its head is numeric unless one child keeps its head symbolic, in which case the row passes that single live parameter on.

Token, token pending. Choose a matching of the three coordinates and solve $T_v = -T_A$ for the carrier $K(F)$: two independent equations in two unknowns, nonsingular because the token map has rank two on the carrier, and the solution expresses $K(F)$ as affine functions of Λ. Writing the two equations as $MK = N\Lambda + c$ with M the carrier block of the token map of v and N that of T_A on Λ, the solution is $K = \operatorname{adj}(M)(N\Lambda + c)/\det M$. Two things have to be said about it, neither of which is automatic. First, dividing by $\det M$ can leave the lattice, and we do not at present have a complete argument that it does not; this is the one step of our construction that we have to record as unfinished, and we do so in Remark B.1 below rather than assert it. Second, a label ℓ of v with carrier row r_ℓ becomes $\ell = -(r_\ell \operatorname{adj}(M))N\Lambda/d + \text{const}$, so its coefficients in Λ are bounded by $2{\cdot}216{\cdot}64 = 27{,}648$ and not by 64; and it is a *non-constant* affine form in Λ, which needs $r_\ell \operatorname{adj}(M)N \neq 0$ rather than merely (E3). We verified this last condition directly, over every ordered pair of a resolving family and a pending coefficient matrix that the catalogue allows and over all matchings of the token coordinates; no label of any family becomes constant. Choose the remaining parameters of v numerically, record the new conditions, and choose Λ in the lattice of Remark B.1 avoiding all conditions in Φ. Each condition is a proper affine subspace, so a lattice point exists in a box of side $M'(|\Phi|+1)$. The coefficient enlargement does not compound: at the end of this step nothing is pending, so the next pending token starts again from catalogue coefficients bounded by 64, and the factor is paid once and not once per owner.

Remark B.1 (the integrality of the eliminated carrier). Solving $MK = N\Lambda + c$ gives $K = \operatorname{adj}(M)(N\Lambda + c)/\det M$, and for the labels of v to be integers this quotient has to be integral. Choosing Λ in a lattice divisible by $\det M$ does *not* achieve this, because the constant c collects the numeric parameters of v and of A that were already fixed and is untouched by the choice of Λ; the family of context `h2__2-6__p2__act` has a token whose minimising carrier has $\det M = -5$

and admits numeric choices for which $\mathrm{adj}(M)c = (384, 336)$, and neither coordinate is divisible by five. Nor does a common lattice follow from a bound on the determinants, since a bound is not a divisibility, and 33 of the determinants that our choice rule selects do not divide 144. We can say the following. The carrier is ours to choose, and when it can be chosen unimodular there is nothing to divide and the difficulty does not arise. Over our catalogue 5,491 of the 7,963 tokens admit a unimodular carrier, and at the level of contexts, which is where the choice is actually made, 2,616 of the 2,789 contexts that occur have a family all of whose tokens admit one. Adding "prefer a family with unimodular carriers" to the choice rule therefore removes the division entirely except at 173 contexts, for which the smallest determinant that can be achieved is at most 36 and lies in $\{2, 3, 4, 6, 8, 9, 24, 36\}$, a set with least common multiple 72. What is still missing is the treatment of those 173. Two routes are available and neither is complete here. One is a single global lattice $D\mathbb{Z}$ with D a common multiple of every determinant that can occur, which makes each individual quotient integral but leaves K in a coarser lattice than the next elimination needs; since the value a cell passes to its parent is its head, and the head is an independent parameter by (E5), this degradation cannot propagate except through a carrier that contains the head or through the rigid rows, so what would be needed is a bound on the length of such a chain. The other is to solve, at each resolution, an affine congruence for the numeric parameters of v, which are chosen at that moment and after $\det M$ is known. That route does not close on its own: the reachable subgroup is generated by $\{M \,\mathrm{adj}(M)v_p \bmod \det M\}$ over the non-carrier columns v_p, and we find that for 2,472 tokens it is a proper subgroup at every admissible carrier, so the congruence is not always solvable. Until the 173 contexts are dealt with, the explicit constant A below is not established, and neither is the integrality of the labels our construction produces in general, although our constructor does produce integer labellings on every tree we have run it on. We are grateful to the referee who supplied the counterexample above.

All labels of A, of its child and parent, and of v become numeric, T_A and T_v are designated antipodes of each other, and nothing is pending.

Parent of the pending cell without a token. As in the first case with a symbolic input. Two-input rows are treated identically with both inputs.

Stationary vertices. A helper places $-x$ (numeric, or the symbolic form $-x$), a two-port absorption places a fresh numeric a and $-x-a$, and neutral units are instantiated with fresh numeric parameters.

Two-token cells. A cell with two tokens pairs its first token with the pending token, if any, exactly as above, and its second token becomes pending. If nothing is pending both tokens become pending with their carriers live. At most two tokens are therefore ever pending, they always belong to a single cell, and the live set has at most four carrier parameters. The bound $3W$ on the number of symbolic labels, and with it the bound on $|\Phi|$, is unaffected, since it counts the cells that carry symbolic labels and not the parameters.

Forced-antipode children. The token $\{-x, a, x-a\}$ of a forced-antipode child stays pending with its carrier live until the parent is processed. The parent's port block or input-in-token family supplies the token $\{x, b, -x-b\}$, and the single equation $a = -b$ is solved for the child's carrier (its token map has rank two on the carrier, so the equation is non-degenerate). A second token of an input-in-token family is an ordinary free token and is treated as above.

Root. The root family contains the zero label. If one token is still pending, choose Λ avoiding Φ as above and hand the negatives of the triple to an odd ordinary block of size at least three. Such a block exists because the special support then has odd size, so the ordinary block sizes have odd sum, and no block has size one. If two tokens T, T' are pending, impose $T' = -T$: two independent equations in four live parameters, solvable because both token maps have rank two on their carriers and their constant parts are zero-sum, and the special support is then closed under negation.

Verification of the invariant and of the bounds

(a) follows from admissibility; (b) from the two token cases; (c) from (E3) and the choice rule, since a symbolic label is one of at most $3W$ labels of three adjacent cells, each a non-constant form in the live coordinates, and two of them are identical or antipodal as forms only if a parent input-only label coincides with a child head-only label, which the choice rule excludes; (d) by construction; (e) because every owner adds at most W placed labels and at most $|\Phi(F)| \leq 3$ reservations. Every box has side $O(|\,\text{Placed}\,| + |\text{Reserved}| + |\Phi|) = O(|S|)$, every label is a bounded-coefficient combination of at most seven parameters, and $|S| \leq 12D + 3$ by Lemma 3.9, and all magnitudes are therefore at most $A(D+1)$ for an absolute A. Explicitly, with $W = 29$, $M = 24$ and $|\,\text{Placed}\,| + |\text{Reserved}| \leq 18(D+1)$, the coordinate boxes have side at most $M\big(2W(18(D+1)+1) + W^2 + 1\big) = 25{,}056(D+1) + 21{,}600$. The pairing boxes use the refined lattice $M' = 3{,}456$ of the elimination step and have side at most $M'\big(3W(36(D+1)+3W)+1\big) = 10{,}824{,}192(D+1) + 26{,}161{,}920$. A label is a sum of at most 7 terms whose coefficients are at most 64 before an elimination and at most 27,648 after one (over a denominator that only decreases the magnitude), so every label has magnitude at most $7 \cdot 27{,}648 \cdot \big(10{,}824{,}192(D+1) + 26{,}161{,}920\big) \leq 8 \cdot 10^{12}(D+1)$. This constant is four orders of magnitude larger than the one a naive reading of the carrier elimination would give, and only the threshold, and not the statement of either theorem, depends on it. At the end every special label other than the pending triple has its antipode placed, so the special support is closed under negation up to that triple. $\square$

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Table 1: The neutral blocks of Lemma 3.7. The labels are read as multiples of a fresh parameter t, the ports first and then the arms in the order listed, each arm from the owner outwards. The equal pairs are the closed form $\pm(-2)^i$; the others come from an exhaustive search and are re-verified from the labels alone.

ports	arms	token	labels
0	1, 1		$1, -2, -1, 2$
0	1, 5		$4, -2, -4, 3, 1, 2, -1, -3$
0	1, 6	yes	$(-2,3), (3,-4), (2,-3), (-1,1), (-3,5), (1,-2), (1,-1), (-4,6), (3,-5)$
0	1, 7		$1, -2, -1, 2, 3, -8, 5, -3, -5, 8$
0	2, 2		$1, -2, 4, -1, 2, -4$
0	3, 2	yes	$(3,-2), (-6,3), (3,-1), (0,-1), (-3,2), (-3,1), (6,-2)$
0	2, 4		$6, 2, -8, -6, 4, -2, 8, -4$
0	2, 5	yes	$(-3,4), (0,1), (3,-5), (3,-4), (-3,5), (1,-2), (-2,3), (-1,1), (2,-3)$
0	2, 6		$5, -2, 4, -5, 3, 2, -3, -1, -4, 1$
0	2, 8		$8, -7, -1, -8, 3, -5, -2, 7, 1, 2, 5, -3$
0	3, 3		$1, -2, 4, -8, -1, 2, -4, 8$
0	3, 4	yes	$(2,0), (-1,-6), (0,6), (-2,-6), (-2,0), (1,-6), (1,6), (-1,0), (2,6)$
0	3, 5		$5, 1, 3, -6, -5, 4, -3, 6, -1, -4$
0	3, 6	yes	$(6,-5), (0,-1), (0,2), (3,-3), (-6,5), (6,-6), (0,1), (-6,4), (3,-1), (-3,3), (-3,1)$
0	3, 7		$1, 5, -6, -2, -1, 3, -8, 2, -5, 6, -3, 8$
0	4, 4		$1, -2, 4, -8, 16, -1, 2, -4, 8, -16$
0	5, 4	yes	$(6,-5), (0,3), (0,-2), (0,1), (0,-4), (0,2), (-6,5), (6,-2), (0,-3), (0,-1), (-6,6)$
0	4, 6		$2, -1, 4, -7, 5, -2, -3, -4, 3, 1, -5, 7$
0	4, 8		$4, 1, -6, 2, -8, -4, 5, -7, 6, -2, -5, 7, -1, 8$
0	5, 5		$1, -2, 4, -8, 16, -32, -1, 2, -4, 8, -16, 32$
0	5, 6	yes	$(5,-4), (-5,5), (0,-2), (-5,3), (5,-1), (0,-3), (-5,4), (0,1), (0,2), (5,-3), (-5,1), (0,3), (5,-6)$
0	5, 7		$2, -3, 4, -1, 6, -8, -2, -4, 8, -6, 3, 5, 1, -5$
0	6, 6		$1, -2, 4, -8, 16, -32, 64, -1, 2, -4, 8, -16, 32, -64$
0	6, 8		$1, 6, -7, 8, -3, 7, -5, -1, -6, 3, 5, -2, -4, 2, 4, -8$
0	7, 7		$1, -2, 4, -8, 16, -32, 64, -128, -1, 2, -4, 8, -16, 32, -64, 128$
0	8, 8		$1, -2, 4, -8, 16, -32, 64, -128, 256, -1, 2, -4, 8, -16, 32, -64, 128, -256$
1	1, 2		$3, -1, 2, -2, 1, -3$
1	1, 4		$6, -8, 4, 2, -4, -2, -6, 8$
1	1, 6		$2, 1, 4, -3, -1, -2, 5, -4, 3, -5$
1	1, 8		$1, -3, 8, 2, -5, 3, -4, -1, 5, -2, 4, -8$
1	3, 2		$3, -2, 1, -3, 5, -1, 2, -5$
1	5, 2		$3, -4, 2, -1, -2, -3, 5, 1, -5, 4$
1	7, 2		$5, -3, 2, -1, -4, 1, 3, -5, 8, -2, 4, -8$
1	3, 4		$5, -2, -1, -4, 2, -3, 4, -5, 1, 3$
1	3, 6		$2, 3, -2, 5, -8, -5, 4, 1, -3, -1, -4, 8$
1	3, 8		$4, 3, -5, -2, 7, -7, 6, -3, 5, 1, -4, -1, 2, -6$
1	4		$3, -3, 1, -2, -1, 2$
1	5, 4		$7, -5, 3, 5, -8, 2, -7, -2, -6, 8, -3, 6$
1	7, 4		$8, -2, 5, -1, -5, 2, 3, -4, 6, -6, 1, -3, 4, -8$
1	5, 6		$1, 2, 3, 5, -6, 4, -8, -3, -2, -1, -5, 8, -4, 6$
1	5, 8		$6, 1, 4, -8, 2, -5, 8, -7, 5, -6, -2, 3, -1, -4, -3, 7$
1	6		$5, -5, 3, -1, 2, 1, -2, -3$
1	7, 6		$7, 1, 4, -5, 3, -1, -7, 2, -6, -8, 5, -4, -3, 6, -2, 8$
1	8		$7, -7, 3, 1, -4, -3, 2, -1, 4, -2$
2	—		$1, -1$
2	5		$5, -3, -2, 1, 2, -1, 3, -5$
2	7		$3, -2, -1, 6, -5, 2, -3, 5, 1, -6$

Table 2: The certificate table, by kind of context: the contexts that occur (Lemma 3.8), those for which the catalogue carries a family, the number of families, the contexts with a family satisfying (E6), and the contexts whose chosen family carries a token. The residue-two rigid row is a constructor template and not a symbolic family, so it is counted as covered in the second column and in none of the others.

context kind	contexts	with a family	families	(E6)	token
bottom cells	954	954	1,563	913	493
continuing cells, free child	419	419	1,777	417	406
continuing cells, joined child	876	876	1,807	810	440
two-input cells	34	34	213	9	33
root cells	507	507	1,332	499	352
total	2,790	2,790	6,692	2,648	1,724